Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2424)

Key

SECTION – I

1.	C	7.	C	13.	D	19.	A
2.	D	8.	B	14.	B	20.	4123
3.	B	9.	B	15.	B	21.	B
4.	D	10.	C	16.	A	22.	2143
5.	B	11.	A	17.	B	23.	D
6.	C	12.	C	18.	B	24.	3214

SECTION – II

1.	246	6.	A	11.	192500	16.	A
2.	D	7.	26	12.	B	17.	D
3.	B	8.	C	13.	D	18.	C
4.	C	9.	3	14.	B	19.	2
5.	B	10.	C	15.	77500	20.	C

SECTION – III

1.	A	7.	2	13.	C	19.	B
2.	22	8.	A	14.	A	20.	C
3.	1	9.	2875	15.	D	21.	B
4.	1	10.	15	16.	C	22.	D
5.	B	11.	B	17.	480		
6.	D	12.	D	18.	C		

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 472

1. Option A: The exploits of the Vikings had positive effects on Europe as well as negative ones, and their accomplishments were essential in shaping the course of European history. Hence choice A is not true.
Option B: The Viking raids began in the ninth century. The second part of choice B cannot be inferred from the passage.
Option C: The Viking warriors swept down from the north like ravenous wolves, spreading terror and bloodshed wherever they went. From their Scandinavian homelands .... And around the turn of the millennium, Viking sailors from Iceland discovered North America and established settlements on its coast – the first Europeans to do so. Hence choice C is true and is the answer.
Option D: Throughout Europe, people trembled in fear at the very mention of their name. .... spreading terror and bloodshed wherever they went. ... The destruction they wrought greatly contributed to the turmoil experienced by western Europe during the Dark Ages. So, choice D is not correct. Choice (C)

2. Refer to the third para.
Option A: Across Scandinavia, the pagan worshipers of Odin and Thor began to convert to Christianity; as good Christians, they became reticent to attack the churches and monasteries that they had once plundered with abandon. While we can infer from the passage that the pagan worshipers of Odin and Thor are the Vikings, choice A is a distortion. It has not been stated in the text that the converted Vikings eschewed **all** forms of violence. They only became reticent to attack the churches and monasteries. Hence choice A is not the answer.
Option B: By the middle of the eleventh century, the glory days of the Viking raiders had faded into twilight, never to return. Hence choice B is wrong.

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Option C: As Scandinavia became increasingly civilized, its kings discouraged the activities of roaming warrior bands, and its social environment gave rise to a more humdrum life. So, choice C is incorrect.

Option D: The growth of strong centralized monarchies in Norway, Denmark, and Sweden also "ultimately resulted in taming the Viking spirit." Therefore, choice D is the answer. Choice (D)

3. Refer to para 3. "As Scandinavia became increasingly civilized, its kings discouraged the activities of roaming warrior bands, and its social environment gave rise to a more humdrum life." By the middle of the eleventh century, the glory days of the Viking raiders had faded into twilight, never to return. Hence, choice B is the correct answer. The remaining choices do not apply in the given context. Choice (B)

4. Option A: The *Greenlanders' Saga*, the earlier of the two was committed to writing in the twelfth century and has about it a primitive crudeness which, while not particularly attractive literally, adds to its historical credibility. The *Eric Saga* has a more polished appearance. So, choice A is incorrect.

   Option B: The earliest written source having this information is not the saga accounts. The earliest written source is *The History of the Archbishops of Hamburg*. So 'Hamburg's saga' as given in choice B is incorrect. 'Flatey's saga' as given in choice B is also incorrect. The Flatey Book, compiled towards the end of the fourteenth century in northern Iceland, contains the earliest extant text of the Greenlander's saga. The Greenlander's saga and the Eric's saga are the only two sagas mentioned in the passage. Hence choice B is not correct.

   Option C: *Greenlanders' Saga*, the earlier of the two was committed to writing in the twelfth century and has about it a primitive crudeness which, while not particularly attractive literally, does add to its historical credibility. The *Eric Saga*, on the other hand, has a more polished appearance. So the first part of choice C is correct. The *Greenlanders' Saga* is more reliable and its text and story more faithful to an oral original. So the second half of choice C is not true.

   Option D: The *Greenlanders' Saga* and the *Eric Saga* tell essentially the same story, yet in some places they complement and in other places they contradict one another. Hence choice D is true. Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Option A: The author mentions that research in AI has delivered in many areas. So we cannot say we have not been able to understand Artificial Intelligence completely. Further, the author does not mention that understanding AI is necessary for building an efficient Artificial General Intelligence.

Option B: According to the author, "we have never actually worked out what natural intelligence is, so we don't know what it is that we are trying to imitate in our machines". In order to understand what natural intelligence is, the author went back "to the source of human intelligence - the human brain and tried to see how we might use computers to better understand this mysterious organ". We can infer from this that according to the author, the reason why we were not able to build an effective Artificial General Intelligence is that we do not understand natural intelligence. To understand natural intelligence, the author studied the human brain.

So, let's see what we have:

Statement – We don't understand natural intelligence

Reason – The brain is the source of natural intelligence, and we haven't understood how that source works.

Step considered – So let's try to understand the working of the brain first.

Therefore –

(i) Until we understand the working of the brain, we won't understand natural intelligence.

(ii) Our primary lack, in the creation of Artificial General Intelligence, is an understanding of how the brain works.

So choice B is the reason that we were not able to build effective Artificial General Intelligence.

Option C: The passage does not talk about the Turing test when it talks about the failure to deliver Artificial General Intelligence. Hence this is not the correct answer.

Option D: The author mentions what the Turing test is in the second paragraph of the passage. Hence, this is not the correct answer.

Therefore, the correct answer is option B.

Choice (B)

6. The passage talks about Alan Turing and his 'Imitation Game' at the beginning of the passage. Turing thought that all that a machine would require to pass his test was more memory - the 1948 Manchester 'Baby' was quite powerful enough already. He estimated that a gigabyte (a thousand million bytes) of memory should suffice, and this should be achievable by the end of the twentieth century.

   By the beginning of the 21st century computers did, indeed, typically have a gigabyte of memory, and they were a million times faster than the

Number of words: 557

5. In the fifth paragraph of the passage, the author states, "research into artificial intelligence (AI) has delivered in many areas of life". But we have failed to build an effective Artificial General Intelligence. 'Baby', but still they could not pass his test. Even today, with still far more computing power and memory, no machine has convincingly passed the test. This would have surprised Turing had he lived to see it.

Option A: According to Turing, "a gigabyte (a thousand million bytes) of memory should suffice" for computers to pass his test. He also predicted that "this should be achievable by the end of the twentieth century". In the next paragraph, the author mentions that "By the beginning of the 21st century computers did, indeed, typically have a gigabyte of memory". Hence, this is not a mistake on the part of Turing.
Option B: The passage does not mention Turing talking about us understanding how our brain functions. Hence, this is not the correct answer. {Even the following - "Can machines think?" He then went on to suggest that this question is difficult to answer directly, and he turned it around into a research experiment that he called 'The Imitation Game', but which subsequent generations simply know as the Turing test for Artificial Intelligence. - is not enough to support choice B.}
Option C: According to Turing, "all that a machine would require to pass his test was more memory". Even though computers with more memory was available by the twentieth century, the machines were not able to pass the Turing test. Hence, his assumption that large amount of memory will be sufficient for a machine to pass his test is not correct. Choice C is the answer.
Option D: Turing did not talk about Artificial Intelligence as such. He only wondered whether machines will be able to think and devised the Turing test. Hence, this is not the correct answer.
Therefore, the correct answer is option C.
Choice (C)

7. The passage talks about Artificial Intelligence and Artificial General Intelligence in the fifth paragraph.
Option A: The passage mentions that by the 21st century, computers "were a million times faster than the 'Baby'". The passage also states that these computers were not able to pass the Turing test. Hence, being fast is not what differentiates AI from Artificial General Intelligence.
Option B: The passage does not mention that because of AI we are able to "talk into our

Option A: The passage states that "the 1948 Manchester 'Baby' was quite powerful enough already". While we can infer from the passage that it was the first of its kind, we cannot infer that it was the most powerful computer. Hence, this is not the correct answer.
Option B: According to Turing, "a gigabyte (a thousand million bytes) of memory should suffice" to pass his test. He also felt that "all that a machine would require to pass his test was more memory - the 1948 Manchester 'Baby' was quite powerful enough already". From this we can infer that the Manchester 'Baby' did not have a gigabyte of memory. If it had, Turing would not have the need to predict that computers with a gigabyte of memory would have passed his test. Hence, this is the correct answer.
Option C: Turing wrote his paper on 'Imitation Game' in Manchester and "it was this machine that brought Turing to Manchester". However, the passage does not explicitly mention whether Turing performed his test on the Manchester 'Baby'. Hence, this cannot be inferred from the passage.
Option D: The passage states that the machines in the 21st century "were a million times faster than the 'Baby'". But it does not mention that the Manchester 'Baby' had a better chance of passing the test.
Therefore, the correct answer is option B.
Choice (B)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 453

9. The answer is stated in the first para "linguistic invention in most subcultures of the modern West is a halting and largely unconscious process. Hackers, by contrast, regard slang formation and use as a game to be played for conscious pleasure." Choice B states this.
Option A: Choice A is contradicted by the sentence: ... the electronic media which knit them

smartphones". Hence, recognizing the meaning of words is not the differentiating factor.
Option C: Artificial General Intelligence is "the idea that a suitably programmed machine might display aspects of intelligence that we normally associate only with humans". Hence, this is the feature that will be in Artificial General Intelligence but not in AI.
Option D: The passage does not mention about the emotional state of a person and understanding it.
Hence, the correct answer is option C.
Choice (C)

8. According to the passage, "the world's first operational electronic stored-program computer ran its first program in Manchester, on June 21, 1948". This was referred to as the Manchester Baby.

together are fluid, 'hot' connections, well adapted to both the dissemination of new slang and the ruthless culling of weak and superannuated specimens.
Option C: Choice C has not been mentioned in the passage.
Option D: Choice D is also wrong. Linguistic invention in most subcultures of the modern West (and not the hacker subculture) is a halting and largely unconscious process. Choice (B)

10. Refer to the last para. But it is also true that hackers use humorous wordplay to make strong, sometimes combative statements about what they feel. Some of these entries reflect the views of opposing sides in disputes that have been genuinely passionate; this is deliberate. We have not tried to moderate or pretty up these disputes; rather we have attempted to ensure that *everyone's* sacred cows get gored, impartially.

Option A: Choice A is contradicted by "We have not tried to moderate or pretty up these disputes ...." Also, choice A does not explain the last sentence of the passage which is given in quotes in the question.
Option B: Hackers use humorous wordplay to make strong, sometimes combative statements about what they feel. Choice B is true but does not answer the question. It does not reflect "impartially" mentioned in the question statement.
Option C: We have not tried to moderate or pretty up these disputes; rather we have attempted to ensure that *everyone's* sacred cows get gored, impartially. Here "we" represents the Jargon File keepers. "these disputes" refers to the disputes of the hackers themselves. It's the hackers who come up with terms which are the 'combatants' with 'sacred cows'. So, it's their sensibilities that are going to be offended without favour (and not those of the readers. The readers are going to find the work interesting and humorous: Even a complete outsider should find at least a chuckle on nearly every page, and much that is amusingly thought-provoking). This makes choice C the correct answer.
Option D: Choice D is true of the slang style which is high-context communication (elliptical, emotive, nuance-filled, multi-modal, heavily coded), which is associated with cultures which value subjectivity, consensus, cooperation, and tradition. But choice D does not answer the question.
Choice (C)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 553

13. Refer to the fourth para. The doctrine as to the ethics of economic conduct, which had been formulated by medieval Popes, was rehearsed by the English divines of the sixteenth century, not merely as the conventional tribute paid by a formal piety to the wisdom of the past, but because the swift changes of the period in commerce and agriculture had, not softened, but **accentuated, the problems of conduct** for which it had been designed. Hence choice D is the answer.
Option A: Choice A provides incomplete facts and can be ruled out.
Option B: The first part of Choice B sounds positive by itself. It is not sufficient to answer the question. The second part of choice B repeats a line from the passage: not merely as the conventional tribute paid by a formal piety to the wisdom of the past. Choice B is not the answer.
Option C: As the changing scenario had aggravated the problem, it naturally follows that it could not solve or do away with the problems of conduct. Choice C is incorrect. Choice (D)

14. Refer to the third para. "The special economic malaise of an age is naturally the obverse of its special qualities." 'obverse' means the counterpart of a fact or truth. The author makes the statement

11. From a careful reading of para 2, we can infer that hackerdom is themed around extremely low-context interaction with computers and exhibits primarily "low-context" values but cultivates an almost absurdly "high-context" slang style. Hence choice A is the answer. Choice (A)

12. Option A: Option A is true and can be understood from the line "The intensity and consciousness of hackish invention make a compilation of hacker slang…this one is the latest version of an evolving compilation called the 'Jargon File'".
Option B: Option B can be considered as true from the lines "Hackers, by contrast, regard slang formation and use as a game to be played for conscious pleasure… combination of the neotenous enjoyment of language-play with the discrimination of educated and powerful intelligence."
Option C: Option C is not true because, from the line "Hacker slang also challenges some common linguistic and anthropological assumptions", we gather that hacker culture cannot be classified by preferred context level unlike other cultures.
Option D: Option D can be understood from the line "Even a complete outsider should find at least a chuckle on nearly every page, and much that is amusingly thought-provoking. But it is also true that hackers use humorous wordplay to make strong, sometimes combative statements about what they feel.". Option D is true and not the answer. Choice (C)

in para 3 (which has been given in the question) and thereafter explains it when he shows how society in medieval England, especially of the common man, was one of people independent in their vocations – whether small farmers, traders or craftsmen – with only a few wage earners. However this very feature meant that there was a great dependence on borrowed funds, since the common folk needed such help to continue with their vocation. Refer to the examples in para 3: The farmer must borrow money ..... The craftsman must buy raw materials on credit .... The young tradesman must scrape together a little capital .... Hence choice B is the answer.
Option A: Choice A is the verbatim translation of the sentence given in quotes in the question. But it does not specifically answer the question about what aspect of society the author is referring to.
Option C: Choice C is too general to describe the discussion which takes place in para 3.
Option D: Virtuosity in choice D means "skill" or "genius". Choice D does not explain the economics of the situation explained in para 3.
Choices A, C and D may draw our attention initially because of the words "reverse", "opposing" and "offset" in them respectively. They are not specific to the question. Choice (B)

15. A distributive state is one where people own resources – natural and manmade. Note: *According to distributists, property ownership is a fundamental right, and the means of production*

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should be spread as widely as possible, rather than being centralized under the control of the state (state socialism), a few individuals (plutocracy), or corporations (corporatocracy). Distributism, therefore, advocates a society marked by widespread property ownership.
Option A: In choice A, the ownership of the capitalists is mentioned. Hence it is incorrect as an answer to the question.
Option B: Tudor England was still a Distributive State. It was a community in which the **ownership of land**, and of the simple tools used in most industries, was not the badge of a class, but the **attribute of a society**, and in which the typical worker was a peasant farmer, a tradesman, or a small master. Hence choice B is correct.

mathematical heft are damnably hard to solve.) and the sentence succeeding blank (2) (And **one such approximation** led Einstein to an odd prediction: any accelerating mass should make ripples in spacetime. Hence (2) is the answer.
The highlighted sentence will disrupt the thought flow if it is placed in blank 1. The sentence "Mass, Einstein realized, deforms the space and time around itself." before blank (1) need to be immediately followed by the sentence: "Gravity is the effect of this, the behaviour of objects dutifully moving along the curves of mass-warped space-time". Hence (1) is not the answer.
The highlighted sentence will be redundant in blanks (3) and (4). The latter half of the para mainly talks about the thoughts and actions of Einstein

Option C: The typical worker was a peasant farmer, a tradesman, or a small master. While this may be true of Tudor England, it is not the reason that 'Tudor England' was called a Distributive State. Hence choice C is not the answer.

Option D: Choice D repeats a fact from para 2. But that does not prove that the state was a distributive one. Choice (B)

16. The term 'usury' is mentioned at the end of para 1: The center of the controversy was the problem which contemporaries described by the word 'usury'. 'Usury' refers to the action or practice of lending money at unreasonably high rates of interest.

Option A: From the last para, it can be inferred that common man could not point out the reason why usury was wrong as it was considered to be a very broad term and had various shades of meaning. Hence choice A is the answer.

Option B: Choice B distorts what has been mentioned in the last para: If the common man was asked why usury was wrong, he would probably have answered with a quotation from scripture.

Option C: While reference has been made to St. Thomas in the penultimate para, there is no mention of St. Raymond's manual. Hence choice C is not the answer.

Option D: Choice D is true from the last sentence of the passage but it does not answer the question. Choice (A)

Solution for question 17:

17. The paragraph begins by discussing the history behind the idea of gravitational waves.

We need to place the highlighted sentence ahead of a sentence which speaks about approximations. The highlighted sentence can be placed only in blank (2). "It is a simple idea, but the equations that give it mathematical heft are damnably hard to solve." in the sentence prior to blank 2 links with "Only by making **certain approximations** can solutions be found**.**" as given in the highlighted sentence. So, the highlighted sentence forms a bridge between the sentence preceding blank (2) (It is a simple idea, but the equations that give it

and all the given sentences in this part of the para need to run continuously. Choice (B)

Solution for question 18:

18. Option A: This option isn't true for two reasons: it makes a grossly incorrect assumption that the present and future are ***hostages to the past***. The para clearly calls the past in debt of the present (an idea in contrast to that of the option). Secondly, the para indicates that a lot can be learnt by analyzing the past. That idea isn't the same as 'people who analyse the past have a lot to gain.' It could be at a societal level too. And being liberated need not be the same as gaining something, it could just be redemption. Hence, Option A can be eliminated.

Option B: This represents the essence as understood from the last line of the para – that analysing the past is more liberating (and hence fruitful) than imagining a future, especially if one manages to uncover the secrets it holds. Hence, Option B is the answer.

Option C: There are distortions in the past, true. There is a lot to learn from the past, also true. But it isn't true that the past is ***distorted by the present.*** Similarly, the para doesn't say the past is unclear ***because of the lies***. The lies it holds is just one of the several things it holds. Hence, Option C is not the answer.

Option D: While the past can be analysed for revelations, the para doesn't really talk about the information being the reason why the past holds an advantage over the future (as if the future won't give us information). The para is more about the past being 'more liberating' that the future, not that there is a tangible competition between past and future. Option D is not the answer. Choice (B)

Solution for question 19:

19. The paragraph states that there are times when managers should innovate and there are times when they should imitate and managers must choose wisely. It does not specifically state that a manager has to focus only on one of these things. Hence, choices (B) and (C) are completely wrong. In choice (D) a part of the statement is true but the second part is an incorrect interpretation.

Choice (A) is closest to the content of the passage and talks about the need for constant evaluation of the competitive environment before choosing either strategic option. Choice (A)

Solution for question 20:

20. Sentence 1 starts with 'they' and therefore, needs a precedent.
Sentence 2 starts with 'they' as well but gives individual examples.
Sentence 3 talks about 'they' transforming cities and nations. But, it has 'even' which means a repetition of what is already happening in a sentence before it.
Sentence 4 is obviously the starting sentence because it talks about 'people' who make a difference, being referred to as 'they' in all the remaining sentences.
1 talks about conviction, while 2 talks about actual action. Also, 2 continues what 1 starts with the emphasis on 'every life' (followed by one smile, one stomach…) 3 adds more change with an 'even'. So, 123 is the block. Ans: (4123)

Solution for question 21:

21. On a cursory reading of the sentences, it can be observed that the paragraph mainly deals with how one can best escape the body (or life) and then talks about the sea.
On a careful reading of the paragraph, it can be inferred that the highlighted sentence does not belong to blank (1). The sentence is completely out of place in blank (1), as it interrupts the flow of thought. "Let us not forget poetry, the touch of someone you love; there is, when you think about it, love itself." in the sentence after blank (1) needs to continue after the sentence preceding blank (1) (There is alcohol, also meditation, the endorphin rush of exercise, the top of the mountain and the freewheel cycle downhill ....) (to escape the body).
The highlighted sentence can be a part of blank (2). "It has to be the sea because the sea holds you up" as mentioned in the highlighted sentence follows the sequence of events mentioned earlier: his pushing of the body to be free of the body, when all you really have to do to escape your sad sack of bones and flesh is take your clothes off, walk into the sea, and splash. By placing the highlighted sentence in blank (2), we also see that the remainder of the text flows well. "It also slaps you and throws you about a bit, requiring your admiration, gratitude and respect. The sea is plentiful and cheap, ... connected to every other beach and rock, ... your brain will open as you swim towards the horizon". So (2) is the answer.
The highlighted sentence cannot be a part of blank (3) because the sea has to be introduced much earlier in the paragraph. It cannot be blank (4) because if the highlighted sentence is placed in blank (4), then there will be a complete distortion of thoughtflow. The pronoun 'it' in the sentence "After five minutes' swimming towards it," would incorrectly refer to the "the sea". The pronoun 'it' needs to refer to " the horizon". The last two sentences of the second para need to flow continuously. Choice (B)

Solution for question 22:

22. Sentence 1 talks about now and the future – where we have to defend art.
Sentence 2 talks about a time prior to art theory when art was understood and not critiqued.
Sentence 3 starts with a positive connector 'indeed' and then follows it up with the importance of getting rid of mechanisms that justify and defend bad art.
Sentence 4 though independent lacks logical context since we wonder why we are talking about quarrelling.
Therefore, sentence 2 is the introductory statement of the paragraph. If we go by the chronology, we understand that 'we are stuck' can only follow the context 'none of us can ever retrieve'. The task of defending art can only be followed by how we defend – we can only quarrel. The 'Indeed' supports the tone of Sentence 4, which talks about quarrel, just as 3 does which talks about our obligation to overthrow any defence of art that is obtuse/insensitive/ onerous. It is important to note that 3 cannot follow 1 since 1 uses 'stuck' and 3 uses 'obligation', and hence, these two cannot be connected with 'indeed'.
Ans: (2143)

Solution for question 23:

23. Choice (A) is wrong, because even though the word 'plagiarising' has been used in the passage, the paragraph's tone is not as accusatory as the word denotes.
The second part of choice (B) is out of scope as it is not mentioned that Baptista found many examples of convergence between music and birdsongs.
The comparison with Shakespeare in the first part of choice C is out of scope.
Choice D covers all the points, while maintaining the tone of the paragraph. It is also the only choice which explicitly mentions that Beethoven was a composer. Choice (D)

Solution for question 24:

24. On a careful reading of the sentences, it can be observed that sentence 3 is a general sentence that begins the paragraph. It introduces the topic: Printing has come a long way since Johannes Gutenberg...The remaining sentences can be arranged keeping the chronological order in mind. Sentence 3 is followed by sentence 2 which mentions other processes 'lithography' and 'screen printing'. Sentence 2 is followed by sentence 1. Laser and inkjet printers arrived in the 1970s. Sentence 4 follows sentence 1. "laser and inkjet printers arrived" in sentence 1 is followed by "3D printers emerged". So, 3214.
Ans: (3214)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	3, 17, 19, 24
Medium	1, 2, 4, 5, 6, 7, 8, 10, 11, 18, 20, 21, 23
Difficult	9, 16
Very Difficult	12, 13, 14, 15, 22

SECTION – II

Solutions for questions 1 to 5:

Given that A paid Rp 102. He could have paid in denominations of 2/3/17.

B paid Rp 357. He could have paid in denominations of 3/7/17.

C paid 399. He could have paid in denominations of 3/7/19.

D paid 238. He could have paid in denominations of 2/7/17.

E paid 147. He could have paid in denominations of 3/7.

Since only C could have paid in denominations of 19, he must have paid in Rp 19. The number of notes that he would have paid will be 21.

If E paid in Rp 7, he must have paid 21 notes. This is not possible. Hence, E paid in Rp 3 notes.

B could not have paid in Rp 17 notes, as he would have to have paid 21 notes. Hence, B must have paid in Rp 7 notes. He must have paid 51 notes. A could not have paid Rp 2 notes as he would have then paid 51 notes. Hence, A must paid in Rp 17 notes. D must have paid in Rp 2 notes.

The following table provides the denomination and the number of notes paid by each of the five persons:

Person	Denomination	Number of Notes
A	17	6
B	7	51
C	19	21
D	2	119
E	3	49

1. The total number of notes that the five friends paid the shopkeeper = 246 Ans: (246)

2. The shopkeeper returned 2 notes to A. None of these two notes can be Rp 17 notes. product that A purchased cannot be Rp 82. Choice (D)

3. The shopkeeper returned 2 notes for the product that A purchased. The price of the product that A purchased can be odd if the shopkeeper returned one odd denomination note and one even denomination note. The price of the product that A purchased can be even if the shopkeeper returned two odd denomination notes.
The shopkeeper returned 4 notes for the product that B purchased. The price of the product that B purchased can be odd if the shopkeeper returned four odd denomination notes. The price of the product that B purchased can be even if the shopkeeper returned three odd denomination notes and one even denomination note.
The shopkeeper returned 1 note for the product that C purchased. The price of the product that C purchased can be odd if the shopkeeper returned one even denomination note. The price of the product can be even if the shopkeeper returned one odd denomination note.
The shopkeeper returned 2 notes for the product that D purchased. The price of the product that D purchased can be odd if the shopkeeper returned one odd denomination note and one even denomination note. However, the shopkeeper cannot return a note of even denomination since there is only one denomination which is even and this person paid using that denomination. Therefore, the price of the product that D purchased must be even.
Hence, only statement IV is definitely true. Choice (B)

4. Since E paid Rp 147 for his product and the shopkeeper returned one note in return, the price of this product can be 145 or 140 or 130 or 128. (It cannot be 144 because E paid using Rp 3 notes.) From the given statements, the statement given in option C is sufficient to determine the price of the product that E purchased. Choice (C)

5. B had Rp 7 notes. Choice (B)

Solutions for questions 6 to 10:

6. Given that D won 16 wars against each of B and C. Hence, D has to win 1 war against A.
Also, A must lose 38 wars in total.
D lost a total of 24 wars and B won a total of 36

If the price of the product was Rp 76, the shopkeeper must have returned Rp 26 to A. This must be done by returning one Rp 19 note and one Rp 7 note. Hence, this is possible.
If the price of the product was Rp 92, the shopkeeper must have returned one Rp 3 note and one Rp 7 note.
If the price of the product was Rp 80, the shopkeeper must have returned one Rp 3 note and one Rp 19 note.
If the price of the product was Rp 82, the shopkeeper must have returned Rp 20 to A. This can be done only if he returns Rp 3 and Rp 17 note. Since this is not possible, the price of the

wars.
Consider that B won 24 wars against D. This implies that B has to win 12 wars against the other countries, C has to win 22 wars and D has to win 1 war. The sum of this is 35 but A has to lose 38 wars against these countries. Hence, B cannot win 24 wars against D.
B can win 21 wars against D. In this case, B has to win 15 wars against the other countries, C has to win 22 wars and D has to win 1 war. The sum of this is 38 and A can lose the 38 wars against these three countries.
Hence, the maximum number of wars that B can win against D is 21. Choice (A)

7. Given that B won 12 wars against each of A, C and D. Hence, A must have lost 26 wars against C and D. To maximize the number of wars that A lost against D, we can consider that A lost all 26 wars against D.
D must have won 7 more wars, C must have won 22 wars and A must have won 25 wars.
One of the ways in which this is possible is if A won 4 wars against B, 9 wars against C and 12 wars against D.
D won 7 wars against C and C won 22 wars against B. Hence, the maximum number of wars that D could have won against A was 26.
Ans: (26)

8. For the given condition to be satisfied, the following must be satisfied:
   - The number of wars won by X must be less than or equal to the number of wars lost by Y
   - The number of wars lost by X must be less than or equal to the number of war won by Y.

   We can observe that these conditions are satisfied for three pairs of countries: (C, B); (C, D) and (D, A). Choice (C)

9. From the above solutions, the given condition is satisfied for three pairs of countries: (C, B); (C, D) and (A, D).
Consider that C won all its wars against B.
In this case B won 8 wars and lost 4 wars against the other countries.
If B lost all 4 wars against D, D must have won 29 more wars which can only be against A. Hence, A must have lost 9 wars but B has won only 8 more wars. Hence, this case is not possible.
If B lost 3 wars against D, D could have lost 30 wars against A, B could have won 8 wars

14 wins and 22 losses remaining and C will have 22 wins and 14 losses remaining.
Hence, B must win the remaining 14 wars against C. Choice (C)

Solutions for questions 11 to 15:

Since the average age of all the persons in an age group in Country A is greater than that of Country B, in the scatter graph, for each interval, the point that is to the right will represent the people from Country A and the point that is to the left will represent the people from Country B.

For example, in the age group 0 – 10, the people from Country A spent ₹50,00,000 in restaurants (i.e., the rightmost point in the age group 0-10), while the people from Country B spent ₹25,00,000.

Similarly, we can determine the amount spent by the persons from each age group for each country. This is presented in the following table:

Age Group	**Amount Spent in Restaurants (in ₹ ‘000)**	
	Country A	**Country B**
0 – 10	5000	2500
11 – 20	5000	25000
21 – 30	30000	15000
31 – 40	10000	35000
41 – 50	30000	20000
51 – 60	35000	50000
61 – 70	55000	60000
71 – 80	45000	25000

against A, A could have won 24 wars against D and 1 war against B. This case is possible and the number of wars that B won against D is 3.

Consider the case that C won all its wars against D. D won 5 wars and lost 2 wars against the remaining countries.

If D won 5 wars against B, B must have lost the remaining 21 wars against A. In this case A must win 4 wars but D lost only 2 wars.

Following a similar logic, if D won 4 wars, A must win 3 wars against D which is not possible.

Hence, D must have won a maximum of 3 wars against B in this case.

Consider the case that D lost all its wars against A. In this case, D cannot win any wars against B. hence, the maximum number of wars that D can win against B is 3. Ans: (3)

10. Given that D won 20 wars and lost 20 wars against A. Hence, among the remaining wars, A won 5 wars and lost 18.

D won 13 wars and lost 4.

To minimize the number of battles that B won against C, let us assume that B won 18 wars against A and 4 wars against D. C would have lost 13 wars against D and one war against A. B would have lost 4 wars against A. Then B will have

81 – 90	15000	5000
91 – 100	2500	10000

11. The total amount spent (in ₹ '000) in restaurants by all the persons of Country A whose age is at least 31 years = 10000 + 30000 + 35000 + 55000 + 45000 + 15000 + 2500 = 192500

Ans: (192500)

12. The total amount spent by all the persons of Country B whose age is at least 31 years and at most 70 years = 165000

The total amount spent by all the persons of Country B whose age is at least 41 years and at most 60 years = 70000

Hence, the total amount spent by all the persons of Country B whose age is between 36 and 65 years must be between 70000 and 165000. From the given options, only II and III fall in this range. Hence, 2 values can be the total amount spent by all the persons of Country B whose age is between 36 and 65 years. Choice (B)

13. The required ratio is the highest for 81 to 90 years. Choice (D)

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14. The amount spent per person in Country B is 25% more than that in Country A for four age groups: 11-20, 31-40, 51-60 and 91-100. Choice (B)

15. The total amount (in ₹ '000) spent in restaurants by all the persons of Country B whose age is at most 40 years = 2500 + 25000 + 15000 + 35000 = 77500 Ans: (77500)

Solutions for questions 16 to 20:

From (i), Kelvin purchased before Gauss, and from (ii), Hertz purchased before Faraday. From (iii), Joule purchased after Hertz and before Kelvin. And from (iv), Faraday purchased after Gauss.

Hence, the order in which they purchased the cats is

would be 3x, 9x and 27x units respectively. It can be seen that, P, Q and R working together can do 13x units in a day while P, Q and S working together can do 31x units in a day. Hence the ratio of times taken to complete the work by the former and latter groups is 31 : 13, P, Q and S take 13 days. Choice (A)

2. The maximum number of regions into which the area inside a circle can be divided by drawing n straight lines

$= \frac{n(n+1)}{2} + 1$

For n = 6, we get the number of regions as 22

Ans: (22)

Hertz, Joule, Kelvin, Gauss, and Faraday.
From (iv), Faraday must have purchased Ragdoll or Abyssinian. But since Hertz has purchased a Ragdoll, Faraday must have purchased Abyssinian.
Kelvin and Joule did not purchase a Siamese cat. Hence, Gauss must have purchased a Siamese cat. Joule and Kelvin must have purchased Persian and Sphynx in any order.
The following table gives the order and the breed of cats they purchased.

Hertz	Joule	Kelvin	Gauss	Faraday
Ragdoll	Persian / Sphynx	Sphynx / Persian	Siamese	Abyssinian

16. Faraday purchased Abyssinian Cat. Choice (A)

17. Faraday purchased a cat immediately after the Siamese cat was purchased. Choice (D)

18. The cat that was purchased last would be the youngest. Hence, the Abyssinian cat was the youngest. Choice (C)

19. They could have purchased the cats in two different ways. Ans: (2)

20. The first person to purchase a cat was Hertz. Choice (C)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	11, 12, 13, 14, 15, 16, 17, 18, 19, 20
Medium	7, 8, 10
Difficult	1, 2, 3, 4, 5, 6, 9
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. From the given information Q is thrice as efficient as P. R is thrice as efficient as Q. S is thrice as efficient as R. If the work done by P in a day is x units, the work done in a day by Q, R and S

3. $x + \frac{1}{x} = 1$

$\Rightarrow x^2 + 1 = x$

$\Rightarrow x^2 = x - 1$

$\therefore x^3 = x(x^2) = x(x - 1) = x^2 - x = x - 1 - x = -1$

$\therefore x^6 = (-1)^2 = 1$

$x^9 = (x^3)^3 = (-1)^3 = -1$

$x^{12} = (x^3)^4 = (-1)^4 = 1$

$x^{12} + x^9 + x^6 + x^3 + 1 = 1 - 1 + 1 - 1 + 1 = 1$

Ans: (1)

4. The last digits of powers of 8 are 8, 4, 2, 6...i.e., a period of 4.
The last digits of powers of 3 are 3, 9, 7, 1...i.e., a period of 4.
The last digit of 258^{35} will be 2 ($\because$ 35 = 4k + 3)
The last digit of 393^{58} will be 9 ($\because$ 58 = 4k + 2)
Hence, the last digit of $258^{35} + 393^{58}$ will be 1.
Ans: (1)

5. The exchange of speeds and directions at the first meeting does not make a difference to the time or place of their second meeting. The two cars start together. By the first meeting (when the 2 cars together cover L, the distance between A and B), Q covers 40 km. By the 2nd meeting, the two cars together cover 3L and the distance covered by Q upto the first meeting plus the distance covered by P between the 1st and 2nd meetings is 120 km. Since the point of the 2nd meeting is 20 km from A, L = 100 km. Choice (B)

6. 3 + 33 + 333 + ... to m terms
3/9 (9 + 99 + 999 + ... to m terms
= 3/9 [(10 – 1) + (100 – 1) + (1000 – 1) + ... to m terms]
= 3/9 [(10 + 100 + 1000 + ... to m terms) – m]

$$= \frac{3}{9}\left(\frac{10(10^m - 1)}{10 - 1} - m\right)$$

$$= \frac{3}{9}\left(\frac{10^{m+1} - 10 - 9m}{9}\right)$$

$$= 1/27(10^{m+1} - 10 - 9m)$$

Choice (D)

7. Let $f(x) = ax^2 + bx + c$
$f(4) = 16a + 4b + c$
$f(1) = a + b + c$
It is given that $f(4) = 4f(1)$
$16a + 4b + c = 4(a + b + c)$
$12a = 3c$
$\Rightarrow c = 4a$
$\therefore$ The product of the roots of the quadratic equation $ax^2 + bx + c = 0$ is $\frac{c}{a} = 4$.

Since 2 is given to be one of the roots of $ax^2 + bx + c = 0$

the other root will be $\frac{4}{2} = 2$.

Alternative Solution:
Let $f(x) = k(x-2)(x-a)$
Given, $k(4-2)(4-a) = 4k(1-2)(1-a)$ [$\because f(4) = 4f(1)$]
$\Rightarrow 8 - 2a = -4 + 4a$
$\Rightarrow a = 2$. Ans: (2)

8. Let the number of marbles with C be x and that with (A + C) be y. Let N be the number of marbles with B.

It is given that, $N = \frac{88}{100}y$ and $N = \frac{112}{100}x$

$\therefore \frac{88}{100}y = \frac{112}{100}x \Rightarrow \frac{22}{25}y = \frac{28}{25}x$, i.e., N must be a multiple of 28 and 22. Now, the only common multiple of 28 and 22 between 1 and 500 is 308.

$\Rightarrow y = \frac{100}{88} \times 308$ i.e., $y = 350$

Hence the total number of marbles with all three of them is 350 + 308 = 658. Choice (A)

9. Total cost of purchasing 10 watches = 20000
For earning 15% profit, he has to sell the watches for 23000.
Since he can sell only 8 watches, price of each watch = ₹2,875. Ans: (2875)

10. Speed of A > Speed of B
On a circular track, starting simultaneously and moving in the same direction, two bodies meet for the first time when the faster body takes a lead of one complete lap over the slower body.

	B	A	Total
Number of laps	5	6	11

Together they complete 11 laps in 15 minutes. When moving in opposite directions, they will meet every time they together complete one lap. Therefore 11th meeting will take place when they together complete 11 laps, i.e., after 15 minutes. Ans: (15)

11. Let the present age of Sumanta and Tamal be $6x$ and $2y$ respectively.

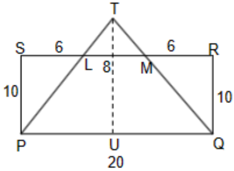
	Sumanta	Tamal
Past	y	x
Present	$6x$	$2y$

Now $y - x = 6x - 2y$
$3y = 7x$
$y = \frac{7}{3}x$
It is given that $y - x = 8$
$\frac{7}{3}x - x = 8$
$\frac{4x}{3} = 8$
$\Rightarrow x = 6$
$\therefore 6x = 36$ and $2y = 2 \times \frac{7}{3}x = 2 \times \frac{7}{3} \times 6 = 28$
Hence, the sum of their present ages is 64 years. Choice (B)

12. Let the initial consumption per day per person be x kg.
Quantity of provision left after 40 days.
$= 240\,(80)(x)$ kg
Let the remaining quantity of food last for d days
$(300)(d)\left(x - \frac{20}{100}x\right) = 240\,(80)x$
$300\,d\,(0.8x) = 240(80)x$
$\Rightarrow d = 80$
$\therefore$ The provisions will get over as per the scheduled time. Choice (D)

13. Let the marked price, the selling price and the cost price of the article be denoted by M, S and C respectively.
It is given that M – S = 3 (S – C) and M = 2C (since the article was marked up by 100% before offering the discount)
$\therefore$ 2C – S = 3S – 3C
$\Rightarrow$ 5C = 4S
$\Rightarrow$ S = 1.25C
Hence, the profit made was 25%. Choice (C)

14.



Let $PQ = 2x$, $QR = x$
Area $= 2x^2 = 200 \Rightarrow x = 10$
$\Rightarrow$ PQ = 20 units and QR = 10 units
Area of triangle SLP = 30 units
Let $SL = y \Rightarrow \frac{1}{2}(10)(y) = 30$ sq.units
$\Rightarrow y = 6$ units
Let the altitude, $TU = h$
Area of triangle TPQ $= \frac{1}{2}(h)(20)$
= Area of triangle TLM + Area of quadrilateral PLMQ
$\frac{1}{2}(h)(20) = \frac{1}{2}(h-10)(8) + \left[\frac{1}{2}(8+20)10\right]$
$\Rightarrow h(20) = (h-10)8 + 280 \Rightarrow 12h = 200$
$h = \frac{200}{12} = \frac{100}{6} = \frac{50}{3}$ units. Choice (A)

15. We know, $225 = 3^2 \times 5^2$

$\Rightarrow$ The power of 3 or 5 has to be 20 or 21 for the highest power of 225 in n! to be 21.

Since, the power of 3 in a factorial is always greater than that of 5, the power of 5 in n! is 20 or 21.

$\Rightarrow$ The number of numbers which have power of 5 as 20 in their factorials is 5(85 to 89, checking using successive division, i.e., 17 + 3 + 0 = 20).

The number of numbers which have power of 5 as 21 in their factorials 5(90 to 94, i.e., 18 + 3 + 0 = 21).

$\therefore$ The required answer is 5 + 5 = 10 Choice (D)

16. $3x + 5y = 24$ and $4x + ky = 32$ will have infinite number of solutions if,

$$\frac{3}{4} = \frac{5}{k} = \frac{24}{32},$$

$$\Rightarrow k = \frac{20}{3} = 6\frac{2}{3}$$ Choice (C)

17. Let the weight of the three pieces be x, $2x$ and $3x$ respectively.

Again $3x$ is cut into 4 pieces which are in the ratio 1 : 2 : 3 : 4

Let the weights of these pieces be denoted by y, $2y$, $3y$ and $4y$ respectively.

$\therefore 10y = 3x$

It is given that $y = 24$, i.e., $3x = 10y = 240$

Weight of the original cake = $6x$ = 480 gm.

Ans: (480)

18. $n^3 - 4n^2 - 16n + 64 \leq 0$

$n^2(n-4) - 16(n-4) \leq 0$

$(n^2 - 16)(n-4) \leq 0$

$(n+4)(n-4)^2 \leq 0$

For all values of n, $(n-4)^2 \geq 0$

Since n is positive $n + 4$ can never be less than zero.

Hence $n = 4$ is the only positive integer value of n that is possible. Choice (C)

19. We observe that, it is an AP series

$a_8 = a + 7d = 46$ and

$a_5 = a + 4d = 28$

solving, we get $d = 6$ and $a = 4$

$\therefore a_{10} = 4 + 9(6) = 58.$ Choice (B)

20. Let O_1 and O_2 be the centres of the circles with radii

Let XY = x

$\therefore$ In ΔO_1PY

$(O_1P)^2 = (PY)^2 + (O_1Y)^2$

$13^2 = (PY)^2 + (O_1X + XY)^2$

or, $13^2 = (PY)^2 + (3 + x)^2$ ------ (1)

In ΔO_2PY

$(O_2P)^2 = (PY)^2 + (O_2Y)^2$

$17^2 = (PY)^2 + (O_2X - XY)^2$

$17^2 = (PY)^2 + (17 - x)^2$ ----- (2)

Equating the expressions for $(PY)^2$ from equation (1) and equation (2), we get

$169 - 9 - 6x - x^2 = 189 - 189 + 34x - x^2$

$40x = 160$ or, $x = 4$

Therefore, XY = 4 and YZ = 6

Now $(PY)^2 = 13^2 - 7^2$ or, PY = $2\sqrt{30}$. Thus PQ = 2(PY) = $4\sqrt{30}$

Area of the triangle with vertices at P and Q inscribed in the overlapping region between the two circles will be maximum when the third vertex is at Z, i.e., it would be an isosceles triangle.

Area of the triangle

$= \frac{1}{2}$ (PQ) (YZ) $= \frac{1}{2}(4\sqrt{30})(6) = 12\sqrt{30}$ sq.cm

Choice (C)

21. Let N = A + B (say A > B)

It is given that $A^3 - B^3 = 133(A - B)$

$\Rightarrow (A - B)(A^2 + B^2 + AB) = 133(A - B)$

$\Rightarrow A^2 + B^2 + AB = 133$ ----- (1)

Again we know that AB = 36

Adding AB to both sides of equation (1), we get

$A^2 + B^2 + AB + AB = 133 + AB$

$A^2 + B^2 + 2AB = 133 + 36$

$(A + B)^2 = 169$

Therefore A + B = $\sqrt{169}$ = 13. Choice (B)

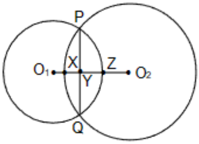
22. $\frac{1}{x-2} > \frac{1}{2x}$

$\Rightarrow \frac{1}{x-2} - \frac{1}{2x} > 0$

$\frac{2x - x + 2}{(x-2)2x} > 0$

$\Rightarrow \frac{x+2}{x(x-2)} > 0$

13 cm and 17 cm.



$O_1Z + O_2X - XZ = O_1O_2$

$13 + 17 - XZ = 20$

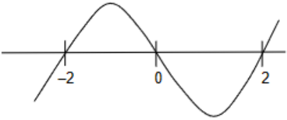
$\therefore XZ = 10$

Thus, $O_1X = 3$ and $O_2Z = 7$

Multiplying both Numerator and Denominator by $x\,(x-2)$, we get $\dfrac{(x+2)(x)(x-2)}{x^2(x-2)^2} > 0$.

The denominator is a positive quantity and so it can be ignored.

$\therefore (x+2)\,(x)\,(x-2) > 0.$



Therefore the solution for the above inequality is $-2 < x < 0$ or $x > 2$

i.e., $x \in (-2, 0) \cup (2, \infty)$

Alternative Solution:

Let's assume x = 1

$\therefore \dfrac{1}{x-2} = -1$ and $\dfrac{1}{2x} = \dfrac{1}{2} \Rightarrow \dfrac{1}{x-2} < \dfrac{1}{2x}$

∴ Option (A) and (B) don't satisfy the given condition.

Assuming x = 3, we get

$\dfrac{1}{x-2} = 1$ and $\dfrac{1}{2x} = \dfrac{1}{6} \Rightarrow \dfrac{1}{x-2} < \dfrac{1}{2x}$.

∴ The only choice satisfying this is (D).

Choice (D)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	9
Easy	2, 4, 13, 16, 17, 19
Medium	1, 5, 7, 8, 10, 11, 12, 15, 18, 20, 21, 22
Difficult	3, 6, 14
Very Difficult	–

Done

Tip: Scroll slowly for the best results